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ALGORITHM & COMPLEXITY

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TOPIC : Digital Watches

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INTRODUCTION

Digital watches are devices that display the time using clear numerical digits, making them easy to read quickly. They are very popular among people of all ages due to their simple design and useful features. Digital watches also provide accurate time display along with additional practical modes. These modes help users manage daily tasks, sports activities, and personal schedules more efficiently.

In this case study, we explore a digital watch that has four main operating modes, which are time display, stopwatch, alarm settings, and timer. Time display mode accurately displays the current hour, minute, and second. Stopwatch mode allows users to measure precise durations, making it useful for activities such as timing events. Alarm Settings mode allows users to set multiple alarms, helping to ensure punctuality and proper time management. Finally, Timer mode allows users to set a countdown, ideal for tasks such as studying, where time is critical.

Understanding how digital watches operate using Finite State Machines (FSM) is important because FSM simplify complex functions into clear steps and states. FSM diagrams help illustrate how these watches seamlessly transition from one mode to another based on user input, without the need for complex programming. Exploring this concept highlights the efficiency and user-friendly nature of digital watches, making it easier for users and developers alike to understand their inner workings and improve their performance.

PROBLEM STATEMENT

In today's era of small and versatile gadgets, digital watches have evolved beyond just showing time. They currently incorporate various operational functions including time display, stopwatch, alarm settings, and timer, all of which demand accurate control logic and smooth user interaction. Balancing these varied functions while maintaining responsiveness and dependability presents a major obstacle in embedded system design.

The main issue is to establish a systematic and dependable control system that regulates the actions of these modes in a digital watch. Every mode should operate autonomously yet also synchronize effectively with others by ensuring seamless state transitions influenced by user actions.

Additionally, the system needs to handle input limitations (e.g., few buttons), provide precise real-time monitoring, and exhibit predictable responses to any user actions, all while avoiding unnecessary complexity in the internal code or interface.

OBJECTIVE

This digital watch case study aims to design a reliable and structured control mechanism using Finite State Machines (FSM). The main goal is to simplify the execution flow of various watch modes while ensuring accurate user interaction, system responsiveness, and efficient transition handling. The proposed system demonstrates how FSM can effectively manage digital watch operations without relying on complex code structures. After analyzing different functional scenarios and user behaviors, the following objectives were identified:

- **Model Digital Watch Behaviors Using FSM/DFA:** Represent and streamline the behavior of various watch modes (Time Display, Stopwatch, Alarm Settings, and Timer) using Deterministic Finite Automata (DFA) to ensure consistent and well-defined state transitions.
- **Define Input-Driven State Transitions:** Clearly identify user actions (such as Start, Pause, Set, Reset) and map them to specific state transitions, ensuring logical operation and predictable system behavior.
- **Develop Visual Diagrams to Illustrate FSM Logic:** Construct detailed flowcharts and DFA diagrams that reflect the internal logic of each mode, helping to visualize and validate system design.
- **Enhance Functional Accuracy Through FSM Design:** Ensure that each mode operates correctly and independently by applying FSM principles, reducing potential bugs or undefined behaviors in the digital watch.

- **Demonstrate Real-World FSM Applications:** Highlight the usefulness of FSM modeling not just in watches but also in broader domains like embedded systems, wearable technology, and consumer electronics

SCOPE

The boundaries of digital watches case study can be defined in terms of three aspects which are user, data and technology used.

User:

1. **General Public:** The primary users of digital watches are typically the general public such as students, workers, and athletes, people who need reliable and easy-to-use time-keeping utilities. These people also known as general consumers will not know how to configure the watches and only uses them for the features provided to them upon purchasing the digital watch.
2. **Developers:** Other groups of users include embedded system developers and computer science students that experiment with the watch such as utilizing FSMs to model digital watch behaviors to fully understand the mechanism behind it. These groups of users are more concerned in finding out how digital watches works rather than how to use said digital watches.
3. **User Interactions:** User interactions of digital watches include pressing the physical buttons or touching the interfaces in the case of smart watches to access functionalities such as time display, stopwatch, alarm settings and timer.

Data:

1. **Input Data:** The traditional digital watches require user input via buttons. Typically, digital watches will have a total of 3 buttons which are START/STOP, MODE, and SET or RESET. The START and STOP button are used for timer mode, MODE button is to change mode while SET or RESET button is for users to manually set the time in the case of the Real-Time Clock (RTC) become faulty. Some version of digital watches will also have a LIGHT button to illuminate the screen of the digital watch in situations where the users are in dark environment.
2. **Output Data:** The output data of digital watch consists of the time and date displayed on the LCD screen. It may also consist of the alarm or beeping sound for notifications.
3. **Stored Data:** Examples of stored data inside digital watches are alarm configurations, current time and date, stopwatch and timer data. Digital watches also save FSM states to know the current mode for the watches.

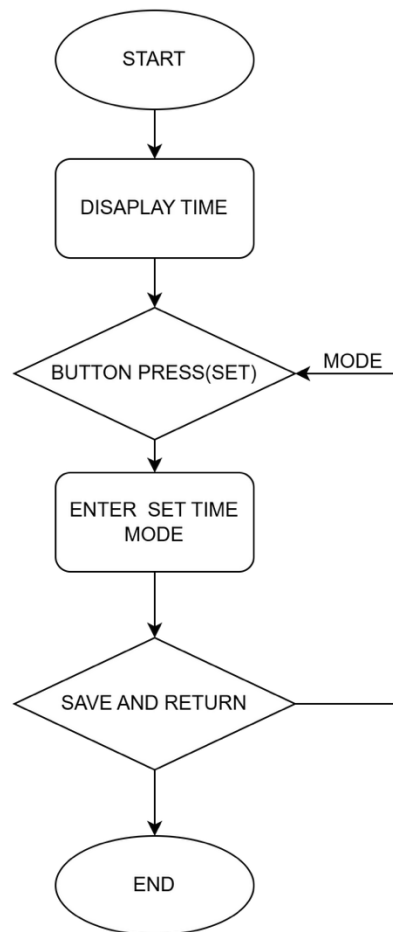
Technology:

1. **Hardware:** Hardwares contained inside digital watches consists of a microcontroller, regulator, RTC system, OLED, LEDs, buzzer sounder, 3-way switch connected to buttons for navigation not including lighting function, power supply like a LiPo battery, and 2 Printed Circuit Boards (PCB) (Kemble, 2013).
2. **Software:** Digital watches contain C language scripts to manage different modes of the watch such as scripts for display time and date, stopwatch, timer, and alarm functions.

MODE: TIME DISPLAY

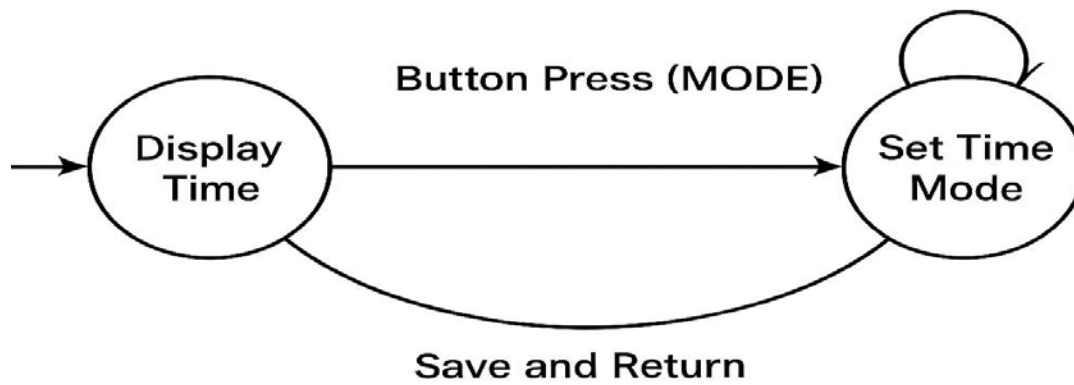
NAME: NUR SABRINA BINTI MOHD SAUPE (CA23011)

FLOWCHART



STATE	DESCRIPTION
START	The digital watch is powered on and initializes to the time display mode
DISPLAY TIME	The current time (in either 12h or 24h format) is shown on the screen
BUTTON PRESS(MODE)	Checks if the user presses the MODE button to switch to other modes
BUTTON PRESS(SET)	Checks if the user wants to set the current time manually
ENTER SET TIME MODE	Enables time adjustment , users can set hours, minutes, and optional date
SAVE AND RETURN	Confirms changes and returns to displaying time
IDLE LOOP	Time continues to update while in display mode unless interrupted by button input

DFA DIAGRAM



List of States (Q):

- A = Display Time
- B = Enter Set Time Mode
- C = Save and Return
- D = Mode Switch

Inputs (Σ):

- 1 = Press MODE button
- 2 = Press SET button
- 3 = Adjust Time (using + / -)
- 4 = Confirm / Save

DFA Explanation:

- $Q = \{A, B, C, D\}$
- $\Sigma = \{1, 2, 3, 4\}$
- $q_0 = A$ (Display Time) , Initial state
- $F = \{A\}$, Final state (returns to Display Time after all operations)

Transition function:

Current state	Input = 1 (mode)	Input = 2 (set)	Input = 3 (adjust)	Input = 4 (confirm)
A (display)	D (mode switch)	B (set time)	A (self loop)	A (self loop)
B (set time)	B (self loop)	B (self loop)	B (adjust time)	C (save and return)
C (save)	A (display)	A (display)	A (display)	A (display)
D (switch)	Switch to next mode	-	-	-

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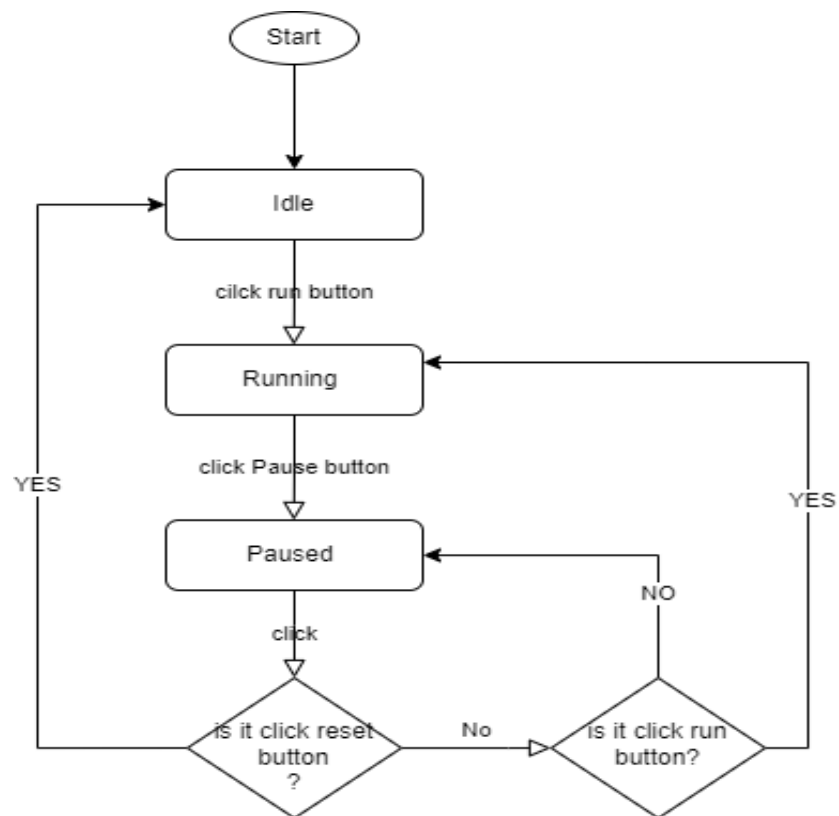
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MODE: STOPWATCH

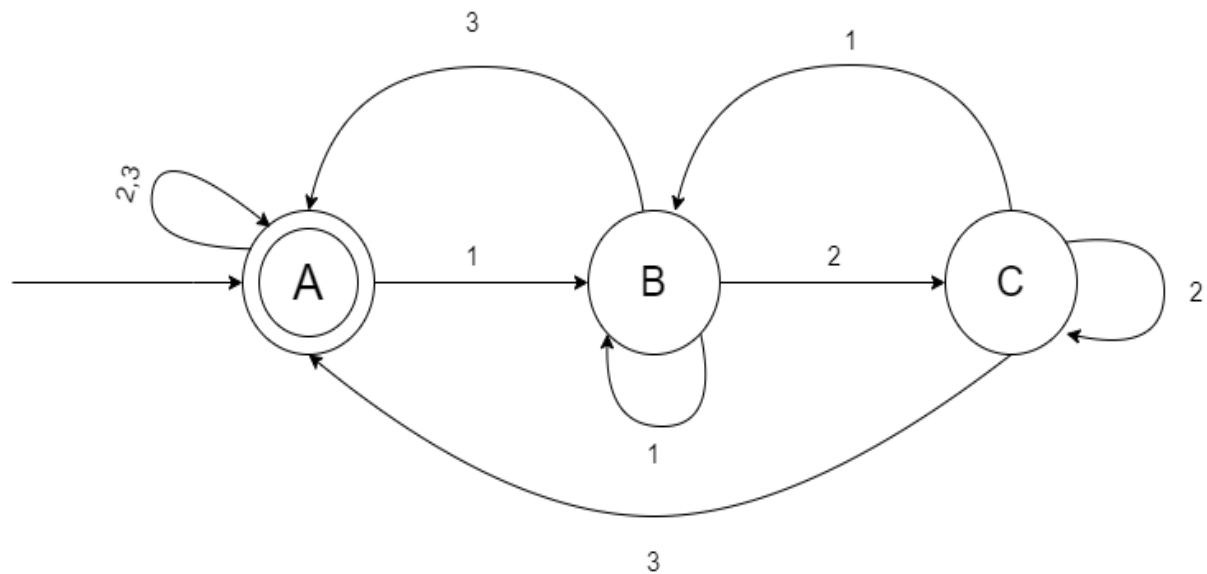
NAME: BAGABER ALI MOHAMMED (CB22017)

FLOWCHART



State / Step	Description
Start	The process begins with the stopwatch in Idle mode.
Idle	The stopwatch is not running. It waits for the user to click the Run button.
Running	The stopwatch is actively counting time.
Paused	The stopwatch is paused and waiting for further action (Reset or Resume).
Click Reset?	A decision point to check if the user clicked the Reset button while paused.
Click Run?	A decision point to check if the user clicked the Run button to resume timing.
Return to Idle	If Reset is clicked, return to Idle.
Return to Running	If Run is clicked from Paused, resume counting.

DFA DIAGRAM



States

- A = Idle
- B = Running
- C = Paused

Inputs (Σ)

- 1 = Click Start/Run
- 2 = Click Pause
- 3 = Click Reset

DFA DIAGRAM EXPLANATION

$$M = (Q, \Sigma, \delta, q_0, F)$$

Where:

- $Q = \{A, B, C\}$
- $\Sigma = \{1, 2, 3\}$
 - o 1: Click Run button
 - o 2: Click Pause button
 - o 3: Click Reset button
- $q_0 = A$ (Idle — initial state)
- $F = \{A\}$ (Final state)
- Transition Function, defined as $\delta: Q \times \Sigma \rightarrow Q$

Current State	Input = 1 (Run)	Input = 2 (Pause)	Input = 3 (Reset)
A: Idle	B (Running)	A	A
B: Running	B (self-loop)	C (Paused)	A (Idle)
C: Paused	B (Running)	C (self-loop)	A (Idle)

- **A: Idle** – Stopwatch is reset and ready to start.
- **B: Running** – Stopwatch is actively counting.
- **C: Paused** – Stopwatch is paused and waiting for resume or reset.

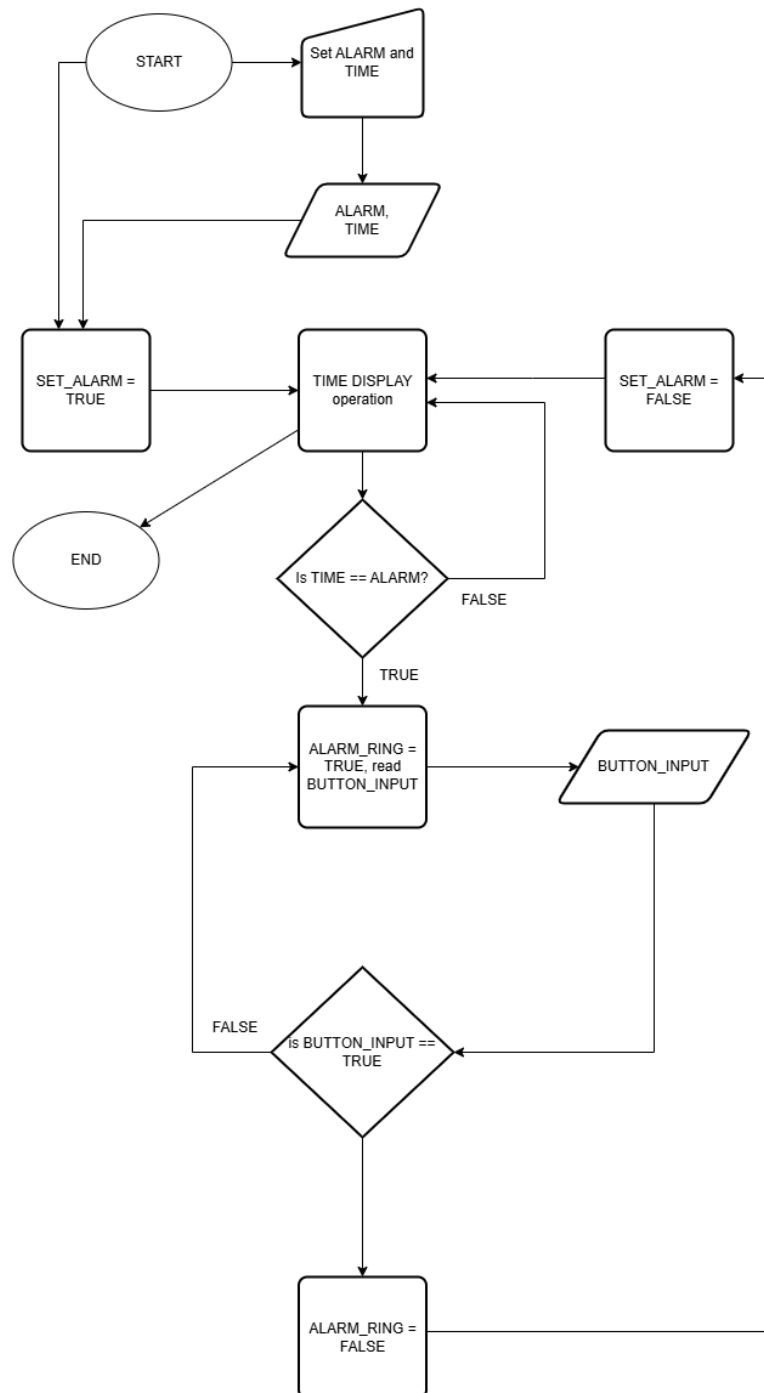
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MODE: Alarm Settings

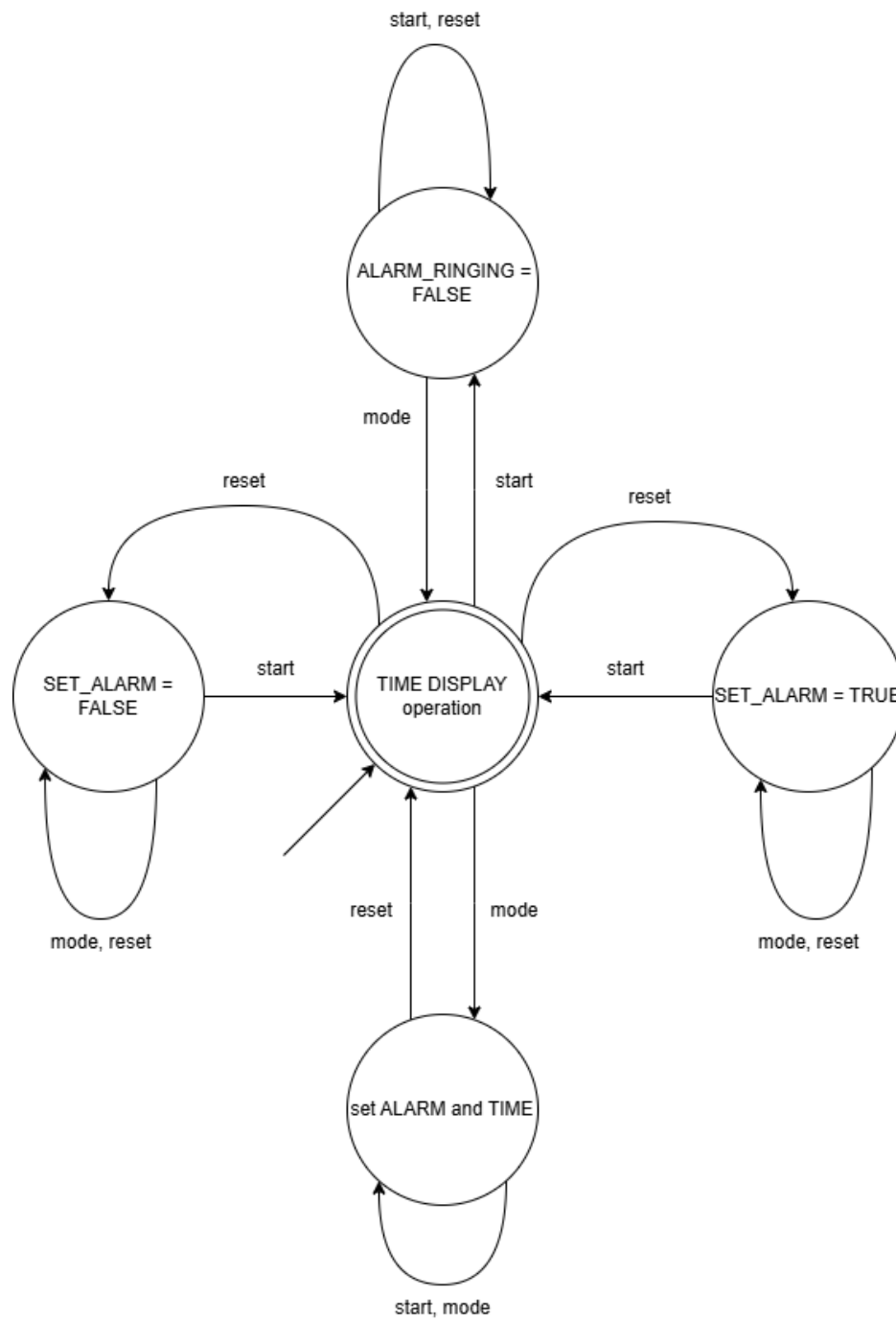
NAME: LIM KAH ZHENG (CD23014)

FLOWCHART



State / Step	Description
Start	The process begins with the alarm either in setting mode or switch mode.
Set ALARM and TIME	The manual input where users must set the alarm time in ALARM and the current time in TIME.
ALARM, TIME	The point where digital watch receives ALARM and TIME as data.
SET_ALARM = TRUE	The process where the alarm is turned on.
TIME DISPLAY operation	The process where the TIME DISPLAY operations are performed.
Is TIME == ALARM?	A decision point to check if the current time is equal to alarm time.
ALARM_RING = TRUE Read BUTTON_INPUT	The process where alarm ringing sound is triggered and the digital watch is prepared to receive any button input from user.
BUTTON_INPUT	The point where digital watch receives any BUTTON_INPUT as data.
Is BUTTON_INPUT = TRUE	A decision point to check if any button input was received from user.
ALARM_RING = FALSE	The process where alarm ringing sound is stopped.
SET_ALARM = FALSE	The process where alarm is turned off.
END	The point where alarm setting ends.

DFA DIAGRAM



States:

- TIME DISPLAY operation
- Set ALARM and TIME
- ALARM_RINGING = FALSE
- SET_ALARM = FALSE
- SET_ALARM = TRUE

Inputs:

- START/STOP button
- MODE button
- RESET button

DFA Diagram Explanation:

$M = (Q, \Sigma, \delta, q_0, F)$

Where:

- $Q = \{\text{TIME DISPLAY operation, set ALARM and TIME, ALARM_RINGING = FALSE, SET_ALARM = FALSE, SET_ALARM = TRUE}\}$
- $\Sigma = \{\text{START/STOP, MODE, RESET}\}$
- $q_0 = (\text{TIME DISPLAY operation})$ — initial state
- $F = \{\text{TIME DISPLAY operation}\}$ — final state
- Transition Function, defined as $\delta: Q \times \Sigma \rightarrow Q$

Current State	Input = START/STOP	Input = MODE	Input = RESET
TIME DISPLAY operation	ALARM_RINGING = FALSE	Set ALARM and TIME	SET_ALARM = TRUE, SET_ALARM = FALSE
Set ALARM and TIME	Set ALARM and TIME (idle)	Set ALARM and TIME (idle)	TIME DISPLAY operation
ALARM_RINGING = FALSE	TIME DISPLAY operation	ALARM_RINGING = FALSE (idle)	ALARM_RINGING = FALSE (idle)
SET_ALARM = FALSE	TIME DISPLAY operation	SET_ALARM = FALSE (idle)	SET_ALARM = FALSE (idle)
SET_ALARM = TRUE	TIME DISPLAY operation	SET_ALARM = TRUE (idle)	SET_ALARM = TRUE (idle)

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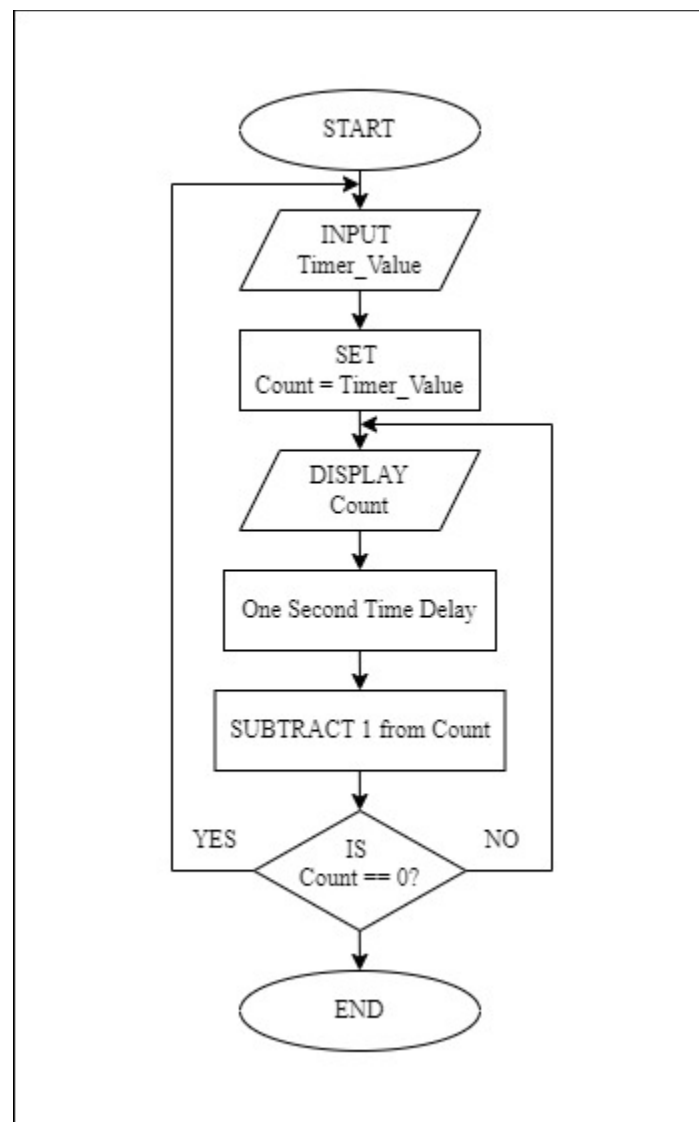
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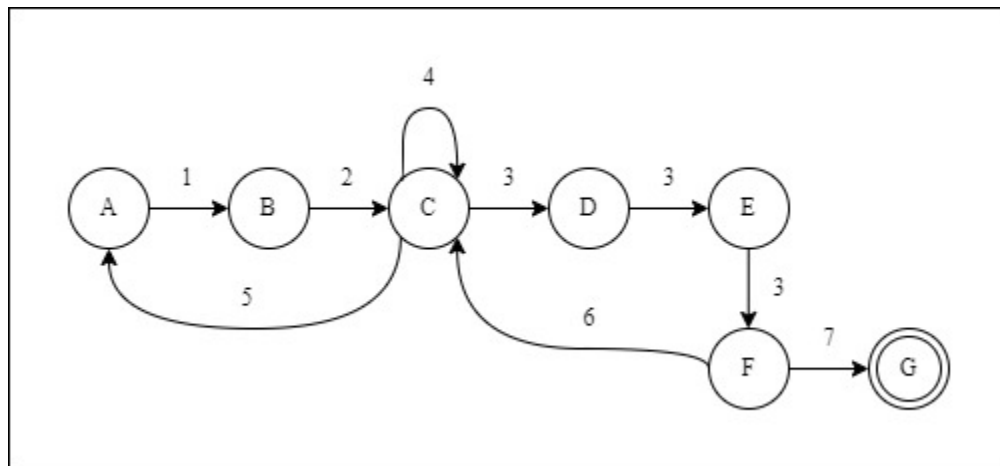
MODE: TIMER

NAME: MUHAMAD ADLI FARRIZ BIN AZLI (CB23086)

FLOWCHART.



DFA DIAGRAM.



$$M = (Q, \Sigma, \delta, q_0, F)$$

Where:-

Q: List of States = {A, B, C, D, E, F, G}

Σ : List of Inputs = {1, 2, 3, 4, 5, 6, 7}

q_0 : Initial States = {A}

F: Final States = {G}

δ : Transition Function = $Q \times \Sigma \rightarrow Q$

i. List of States.

STATES	DESCRIPTION
A	Input Time Value. The user enters the desired countdown duration, such as 30 seconds, using the interface.
B	Set Timer Value. The system sets and stores the countdown value based on the user's input.
C	Display Timer. The timer value is shown on the screen and updated in real-time during countdown.
D	One Second Time Delay. The system waits for one second before proceeding to the next countdown step.
E	Subtract 1. The system decreases the current timer value by one second.
F	Check Condition. The system checks whether the timer has reached zero or should continue counting.
G	Finish. When the timer reaches zero, the system triggers an alert or message to indicate time is up.

ii. List of Inputs.

STATES	DESCRIPTION
1	Set
2	Start
3	Running
4	Pause
5	Stop
6	Continue
7	Finish

iii. Transition Function.

	1	2	3	4	5	6	7
A	B	-	-	-	-	-	-
B	-	C	-	-	-	-	-
C	-	-	D	C	A	-	-
D	-	-	E	-	-	-	-
E	-	-	F	-	-	-	-
F	-	-	-	-	-	C	G
G	-	-	-	-	-	-	-

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